RAJARAJAN T P

+91 7397371356 | rajatp@bu.edu | www.linkedin.com/in/rajarajantp001

EDUCATION

Boston University Boston, MA
MS in Biomedical Engineering (Incoming)

St. Peter's College of Engineering and Technology, Anna University Chennai, India
Bachelor of Technology in Biotechnology (Honors) Nov, 2021 - June, 2025
CGPA: 9.5/10 [3.8/4] till 7th Semester | Gate Biotechnology (2025): 38

Relevant Coursework:

- Biomedical Engineering	- Bioinformatics	- Analytical Instrumentations for Biotechnology	- Material Science for Biotechnologists
- Genetic Engineering	- Cancer Biology	- Statistics and Numerical Methods	- Immunology and Microbiology
- Tissue Engineering	- Molecular and Cell Biology	- Molecular Therapeutics and Diagnostics	- Bioprocess Principles and Engineering
- Protein Engineering	- Down Stream Processing	- Industrial Safety and Total Quality Management	- Basic Electrical, Electronics and Instrumentation Engineering

EXPERIENCE

1. Indian Council of Medical Research - NIRT Chennai, Tamil Nadu
Project Dissertation (Virology Dept.) (1st Jan 2025 to 31st July 2025)

- Conducted in-silico structural and functional analysis of HIV-1 Rev protein and Rev Response Element (RRE) to explore subtype – specific and strain - specific structural and functional variations.
- Maintained HEK293 and Jurkat cell lines and performed viral infection, cloning, plasmid extraction and transformation, PCR techniques by assisting in a project evaluating the functional difference in LTR and TAT regions of HIV-1 and HIV-2.

2. Biofuels Laboratory, Centre for Food Technology Chennai, Tamil Nadu
Research Intern (Algal Research) (July 19, 2023, to May 31, 2025)

- Independently executed the project “Subtractive Proteomics and Virtual Screening of Algal-Derived Compounds Against Diabetes Gangrene-Associated Bacteria”, using Bioinformatics pipeline, docking and in-silico drug discovery using 15 compounds.
- Contributed in data analysis, research and scientific writing for publication of the project “Evaluation of Marine Sulphated Polysaccharides as G6PC1 Inhibitors – A Computational Approach”.

3. Council of Scientific and Industrial Research Chennai, Tamil Nadu
Molecular Biology Intern (NEERI Dept.) (1st July 2024 to 5th Aug 2024)

- Analyzed metagenome sequences using bioinformatics tools and developed phylogenetic tree, heat mapping and operated basic molecular biology lab equipment like Gel Electrophoresis, PCR, etc., using unknown bacterial DNA.

4. Adyar Cancer Institute (WIA) Chennai, Tamil Nadu
Clinical Biochemistry Intern (Biochemistry Dept.) (17th July 2023 to 4th Aug 2023)

- Analyzed various biomarkers and molecules present in the Patient's Blood samples and performed serological works for cancer confirmation.
- Performed serum analysis with advanced equipment such as semi-auto analysers, Gel-Electrophoresis Analyzers, nephelometers, and HPLC

PROJECTS

1. Subtractive proteomics and virtual screening of algal derived compounds against Diabetes Gangrene associated bacteria - A computational approach (Team Size: 1)

- Developed a subtractive proteomics pipeline to predict best druggable bacterial proteins within 4 most persistent bacteria present in Diabetic Gangrene.
- Performed structural and functional in-silico analysis on the bacterial proteins and performed molecular docking to virtually screen algal polysaccharides and identify most suitable drug candidate

2. Comparison of Rev-RRE interactions in HIV-1 Subtype B & C (Team Size: 2)

- Conducted a comparative in-silico analysis of Rev and RRE in HIV-1 Subtypes B and C through evaluation of Sequence, Structural and Interaction profile of 16 Rev-RRE complexes.

- Identified key patterns in variations based on subtype-specific and strain-specific interactions by analysing 3 strains each in HIV-1 subtype B and C.

3. Cardiovascular Healthcare Segmentation: Predictive Analysis Using Machine Learning (Team Size: 1)

- Built a Random Forest model to predict the risk of cardiovascular disease in patients using clinical and demographic data of 100 patients.

4. Unlocking the Potential of Hematococcus pluvialis for Beta Carotene Production through Stress Optimization (Team Size: 2)

- Explored 10 stressors and their effects in beta-carotene productivity using HPLC and optimized the stressors to determine the best combination for enhanced beta-carotene and subsequently Astaxanthin yield using Central Composite Design and RSM.

RESEARCH PUBLICATION

1. **Apoptosis inducing anti-proliferative activity of Citrullus lanatus seeds against A549 cell lines.** South African Journal of Botany, 171, 96-105.
2. **Growth and Production of the Milky White Medicinal Mushroom Calocybe indica (Agaricomycetes) through Varying Ratios of Organic Waste Substrates for Medicinal Applications**. International Journal of Medicinal Mushrooms, 27(10).
3. **Evaluation of marine algal sulphated polysaccharides as Glucose 6 phosphatase catalytic subunit 1 (G6PC1) inhibitors-A computational approach**. Indian Journal of Biochemistry and Biophysics (IJBB), 62(8), 896-908.

SKILLS

Technical	:	Cloning, Plasmid Isolation and Transformation, Cell Culturing (HEK293 and Jurkat Cell Lines), PCR, SDS – PAGE, Chromatography (HPLC, Gel and Affinity Chromatography), Anti-Microbial, Anti-Oxidant and Cytotoxicity Assays, In-silico Drug-Discovery
Software	:	Python, WEKA, CATIA, BIOVIA, Microsoft Office
Professional	:	Leadership, Teamwork, Time management, Multi-Tasking, Adaptability, Critical thinking and Problem Solving

POSITIONS OF RESPONSIBILITY

- Treasurer of UYIRI Association (2024-2025) | Department of Biotechnology., SPCET
- President (2024-2025) and Vice President (2023-2024) of BRSI | Biotechnology Dept., SPCET
- Member of Center for Excellence in Algal Research (CEAR) | SPCET
- Member, YMCA Chennai (2014 – Till Date)
- Active member of St. Peter's College NSS squad

ACHIEVEMENTS

- Best Outgoing Student Award (Biotechnology) | SPCET
- Best Project Award (Biotechnology Department) | SPCET.
- First place for highest CGPA in first 6 semesters.
- Best Paper Award at HAPTEN National Conference (2023).